Econometrics I

Lecture 1: Probability and Statistics

Fall 2026

Roadmap

1. Probability spaces and random variables
2. Distribution and density functions
3. The algebra of expectations: $\mathbb{E}[\cdot]$, $Var[\cdot]$, and $Cov[\cdot, \cdot]$ as operators
4. Multivariate distributions and independence
5. Conditional probability and Bayes's Theorem
6. Conditional expectations, iterated expectations, total variance
7. Matrix notation (details in the self-study handout)

One theme for today: most of this course is applying a small set of rules for expectations and variances, over and over. Learn to push the operators around and the rest of the semester is bookkeeping.

To discuss probability we first need a few basic definitions:

- An **outcome** is something we *can* observe but may not know in advance

  Example: For a coin flip, H (heads) is an outcome

  Example: The wage of a randomly sampled worker

- A **sample space**, Ω, is a set of all possible outcomes

  Example: For a coin flip, $\{H, T\}$ is the sample space

  Example: For two coin flips, $\{HH, HT, TH, TT\}$ is the sample space

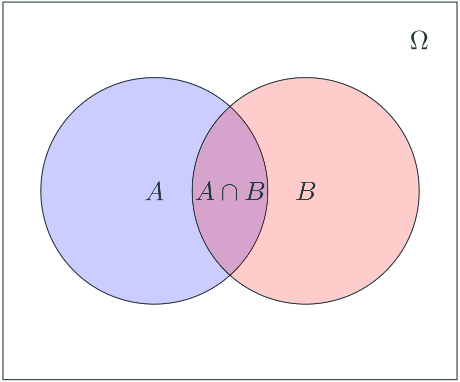
- An **event** is any subset of the sample space

  Example: For two coin flips, $\{HH, TT\}$ is the event "getting the same side both times"

- A **probability** is a function from S, the set of all events, to $[0, 1]$ such that

  1. $P(E) \in [0, 1]$ for any event, E
  2. $P(\Omega) = 1$
  3. $P(A \cup B) = P(A) + P(B)$ whenever $A \cap B = \emptyset$

Outcomes and Events as a Venn Diagram



- A and B are events in the sample space, Ω
- The intersection, $A \cap B$, is the purple part
- The union, $A \cup B$, would be everything that isn't white

- A random variable assigns numeric values to outcomes:

$$X : \Omega \to \mathbb{R}$$

- We can define a probability distribution on this random variable in terms of our original probability space:

$$\mathbb{P}(X = x) = \mathbb{P}\left(\bigcup_{\omega : X(\omega) = x} \omega\right)$$

- *Example:* If rolling two dice, probability of the sum being 11 is given by:

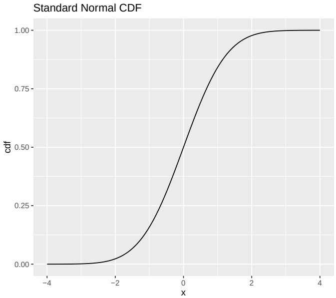
$$\begin{aligned} P(D_1 + D_2 = 11) = P((D_1 = 5, D_2 = 6) \cup (D_1 = 6, D_2 = 5)) = \\ = P(D_1 = 5, D_2 = 6) + P(D_1 = 6, D_2 = 5)) \\ = 1/36 + 1/36 = 1/18 \end{aligned}$$

- For a continuous random variable X the probability that $X = x$ is "0" since any one outcome happens with vanishingly small probability
- Instead we think about *sets* like $\mathbb{P}(a < X < b)$
- Define the **Cumulative Distribution Function** of X to be:

$$F(x) = \mathbb{P}(\omega : X(\omega) \leq x)$$

- **n.b.**: CDF's are always weakly increasing and live in $[0, 1]$
- Actually, any non-decreasing, right-continuous function F with $\lim_{x \to -\infty} F(x) = 0$ and $\lim_{x \to \infty} F(x) = 1$ is a CDF.

Standard Normal (Gaussian) CDF

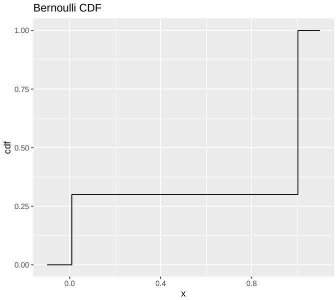



```r
if(!(require(ggplot2))){install.packages('ggplot2')}
ggplot(data.frame(x = c(-4, 4)), aes(x = x)) +
stat_function(fun = pnorm) +
ylab("cdf") +
ggtitle("Standard Normal CDF")
```

$$F(x) = \frac{1}{2}\left[1 + \text{erf}\left(\frac{x-\mu}{\sigma\sqrt{2}}\right)\right]$$

n.b.: $\mu = 0$, $\sigma = 1$, plotted here, is what we call the **standard** normal

Bernoulli CDF




```r
if(!(require(ggplot2))){install.packages('ggplot2')}
sfun0  <- stepfun(0:1, c(0., .3, 1.), f = 0)
x = seq(-.1, 1.1, length.out = 100)
df = data.frame(x = x, y = sfun0(x))
ggplot(df, aes(x,y)) + geom_step() +
ylab("cdf")  +
ggtitle("Bernoulli CDF")
```

The Bernoulli distribution describes a random variable $X \in \{0, 1\}$.

Plotted is Bernoulli distribution with probability of success $\mathbb{P}(X = 1) = p = .7$

CDF: draw your own!

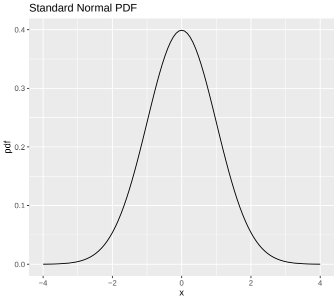
Probability Density Functions

- A random variable's **Probability Density Function** is the derivative of its CDF:

$$f(x) = \frac{d}{dx} F(x)$$

- PDFs are well-defined when the CDF is **absolutely continuous** (which requires continuity and almost-everywhere differentiability)

Standard Normal (Gaussian) PDF




```r
if(!(require(ggplot2))){install.packages('ggplot2')}
ggplot(data.frame(x = c(-4, 4)), aes(x = x)) +
stat_function(fun = dnorm) +
ylab("pdf") +
ggtitle("Standard Normal PDF")
```

$$f\left(x\right) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]$$

n.b.: $\mu = 0$, $\sigma = 1$, plotted here, is what we call the **standard** normal

Discrete vs continuous random variables

- Note that for discrete distributions like the Bernoulli distribution, we have discontinuous jumps in the CDF, and the PDF is not defined.
- Instead of a PDF, we can define a **probability mass function**:

$$p(x) = \mathbb{P}(X = x)$$

- The set of points x such that $\mathbb{P}(X = x) > 0$ are call the **support** of X. Similarly, the support of a continuously distributed random variable can be defined as those points x where the pdf $f(x) > 0$.

The Algebra of Expectations

- For a continuously distributed random variable:

$$\mathbb{E}_F[X] \equiv \int x f(x) dx$$

- For a discretely distributed random variable:

$$\mathbb{E}_p[X] \equiv \sum_{x \in Supp(X)} x p(x)$$

- These expectations are also called the **mean** or **first moment** of X. Often, $\mu \equiv \mathbb{E}[X]$.
- The F subscript denotes the distribution used to take the expectation. We will often omit it when there is no ambiguity.

Expectations of functions of random variables

- For a continuously distributed random variable:

$$\mathbb{E}_F\left[g\left(X\right)\right] \equiv \int g\left(x\right) f(x)dx$$

- For a discretely distributed random variable:

$$\mathbb{E}_p\left[g\left(X\right)\right] \equiv \sum_{x \in Supp(X)} g\left(x\right) p\left(x\right)$$

- Note that functions of random variables are random variables themselves, so we're not adding much here.

$\mathbb{E}[\cdot]$ is an operator: learn its rules

Treat $\mathbb{E}[\cdot]$ like you treat the derivative: an operator with a short list of rules.

- Linearity (constants a, b; any random variables X, Y):

$$\mathbb{E}[a + bX] = a + b\,\mathbb{E}[X] \qquad \mathbb{E}[aX + bY] = a\,\mathbb{E}[X] + b\,\mathbb{E}[Y]$$

  This holds *always* — no independence required.

- Constants pass through; sums split; that's it.

- What linearity does NOT give you:

$$\mathbb{E}[XY] \neq \mathbb{E}[X]\,\mathbb{E}[Y] \qquad \mathbb{E}[g(X)] \neq g(\mathbb{E}[X])$$

  (the first gap is covariance; the second is Jensen's inequality)

Nearly every derivation this semester is: apply linearity, track the pieces that don't split.

Jensen's Inequality

If g is convex, $\mathbb{E}\left[g\left(X\right)\right] \geq g\left(\mathbb{E}\left[X\right]\right)$.

If g is concave, $\mathbb{E}\left[g\left(X\right)\right] \leq g\left(\mathbb{E}\left[X\right]\right)$.

Application to even moments: If $g(x) = x^{2k}$, and X is a random variable, then g is convex as

$$\frac{d^2 g}{dx^2}(x) = 2k(2k-1)x^{2k-2} \geq 0 \quad \forall x \in \mathbb{R}$$

and so

$$g(\mathbb{E}[X]) = (\mathbb{E}[X])^{2k} \leq \mathbb{E}\left[X^{2k}\right]$$

What does this mean if $k = 1$?

Jensen's Inequality

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$$g(\mathbb{E}[X]) = (\mathbb{E}[X])^{2k} \leq \mathbb{E}\left[X^{2k}\right]$$

What does this mean if $k = 1$?

- The variance of a random variable:

$$Var\left[X\right] \equiv \mathbb{E}\left[\left(X-\mu\right)^2\right]$$

where $\mu = \mathbb{E}\left[X\right]$

- Usually, $\sigma^2 \equiv Var\left[X\right]$
- $\mathbb{E}\left[X^k\right]$ is called the k**th (uncentered) moment** of X
- Note that knowing first two moments is equivalent to knowing mean and variance.

- Pareto distribution:

$$F(x) = \begin{cases} 0 & \text{if } x < 1 \\ 1 - \left(\frac{1}{x}\right)^2 & \text{if } x \geq 1 \end{cases}$$

- What are the mean and variance of this distribution?

- Econometric models are usually concerned with several random variables.
- Denote a **collection** of random variables:

$$\{X_T\}_{t=1}^{T}.$$

  Note that t here does not necessarily have anything to do with time.
- Examples:
  - One of the variables is “dependent” (denoted Y) and the others “explanatory”
  - A measurement observed repeatedly over time (a **time series**), e.g.,
    - the price of a stock
  - A measurement observed for different individuals (a **cross section**), e.g.,
    - wages or income
  - Most commonly, we will have a combination of these things: multiple observations of several variables. (**panel data**)

Random vector

- We can now define

$$\mathbf{X}_T : \Omega \to \mathbb{R}^T,$$

where $\mathbf{X}_T\left(\omega\right) = \left(X_1\left(\omega\right), \dots, X_T\left(\omega\right)\right)$

- The **joint** CDF is a natural extension to the CDF for a single random variable:

$$F_{\mathbf{x}}\left(\mathbf{x}\right) = \mathbb{P}\left(\omega : X_1\left(\omega\right) \leq x_1, \dots, X_T\left(\omega\right) \leq x_T\right)$$

  where $\mathbf{x} = \left(x_1, \dots, x_T\right) \in \mathbb{R}^T$.

- The **joint PDF** can be defined as $f\left(\mathbf{x}\right) = \frac{\partial^T F_{\mathbf{x}}(\mathbf{x})}{\partial x_1 \partial x_2 \dots \partial x_T}$.

- **Marginal distributions** refer to the CDF of the individual X_t variables,

$$F_{X_t}\left(x\right) = \mathbb{P}\left(\omega : X_t\left(\omega\right) \leq x\right)$$

  Formally, marginal CDFs can be obtained from joint distributions by integrating over the other variables.

- Covariance is an important property of the joint distribution of random variables:

$$Cov\left(X_1, X_2\right) = \mathbb{E}\left[\left(X_1 - \mathbb{E}\left[X_1\right]\right)\left(X_2 - \mathbb{E}\left[X_2\right]\right)\right]$$

- And more generally for a random vector

$$Cov\left(\mathbf{X}_T\right) = \mathbb{E}\left[\left(\mathbf{X}_T - \mathbb{E}\left[\mathbf{X}_T\right]\right)\left(\mathbf{X}_T - \mathbb{E}\left[\mathbf{X}_T\right]\right)'\right],$$

  which is a $T \times T$ matrix.
- Note that $Cov\left(X, X\right) = Var\left(X\right)$.

Properties of expectations and covariance I

- For constants a, b, you should be able to show that

$$\mathbb{E}\left[a + bX\right] = a + b \cdot \mathbb{E}\left[X\right]$$

- From this, it follows that

$$Cov\left[a + b \cdot X_1, X_2\right] = b \cdot Cov\left[X_1, X_2\right]$$

From this, it follows that

$$Var\left(a + b \cdot X\right) = b^2 \cdot Var\left(X\right)$$

- For constants a, b, you should be able to show that

$$\mathbb{E}\left[aX + bY\right] = a \cdot \mathbb{E}\left[X\right] + b \cdot \mathbb{E}\left[Y\right]$$

- From this, it follows that

$$Cov\left[aX + bY, Z\right] = a \cdot Cov\left[X, Z\right] + b \cdot Cov\left[Y, Z\right]$$

Variances of sums

Two more rules, both provable with linearity alone:

- For any X, Y:

$$Var\,(X + Y) = Var\,(X) + Var\,(Y) + 2\,Cov\,(X, Y)$$

- For $X_1, \ldots, X_n$ uncorrelated with common variance σ^2:

$$Var\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} Var\,(X_i) = n\sigma^2$$

Cashing in: the variance of a sample mean

Let $X_1, \ldots, X_n$ be uncorrelated draws with mean μ and variance σ^2, and let $\overline{X}_n = \frac{1}{n}\sum_{i=1}^n X_i$.
Using only the operator rules:

$$\mathbb{E}\left[\overline{X}_n\right] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\left[X_i\right] = \mu \qquad \text{(linearity)}$$

$$Var\left(\overline{X}_n\right) = \frac{1}{n^2} Var\left(\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\cdot n\sigma^2 = \frac{\sigma^2}{n} \qquad (Var(bX) = b^2 Var(X))$$

- No distributions, no integrals: just the rules.
- $\sqrt{\sigma^2/n}$ is the standard error of the mean — our first standard error.
- Later today we simulate this and watch σ^2/n show up in the data.

- Two events $A, B \subset \Omega$ are **independent** iff

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\, \mathbb{P}(B)$$

- A collection of random variables **X** is **independent** iff

$$F_{\mathbf{x}}(\mathbf{x}) = F_{X_1}(x_1)\, F_{X_2}(x_2) \ldots F_{X_T}(x_T)$$

- A collection of random variables **X** is **independent** iff for any (measurable) real-valued functions $g_1, g_2, \cdots, g_T$,

$$\mathbb{E}\left(g_1(X_1)\, g_2(X_2) \ldots g_T(X_T)\right) = \mathbb{E}\left[g_1(X_1)\right] \mathbb{E}\left[g_2(X_2)\right] \ldots \mathbb{E}\left[g_T(X_T)\right]$$

- What does this imply about the relationship between $\mathbb{E}[XY]$ and $\mathbb{E}[X]\,\mathbb{E}[Y]$?

- The **conditional probability** of event A given event B can be defined as

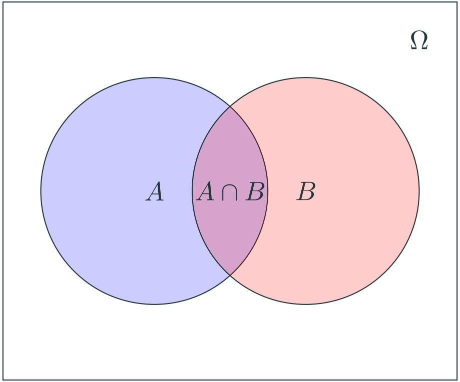
$$\mathbb{P}\left(A|B\right) = \frac{\mathbb{P}\left(A \cap B\right)}{\mathbb{P}(B)}$$

- The **conditional CDF** of random variable X conditional on $Y = y$ can be written

$$F_X\left(x_0|Y = y\right) = \int_{-\infty}^{x_0} \frac{f\left(x, y\right)}{f_Y\left(y\right)} dx$$

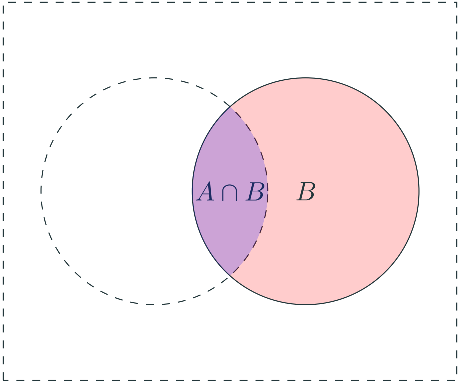
where $f_Y\left(y\right)$ is the marginal density of y

Conditional Probability in a Venn Diagram



- The probability of A, $\mathbb{P}\left(A\right)$ is the relative area of A (within Ω)
- But what if B definitely occurred?

Conditional Probability in a Venn Diagram



- The probability of A is STILL the relative area of A
- But we only take into account the part of A "inside" B

- From $\mathbb{P}\left(A|B\right) = \frac{\mathbb{P}(A\cap B)}{\mathbb{P}(B)}$, we have

$$\mathbb{P}\left(A\cap B\right) = \mathbb{P}\left(A|B\right)\mathbb{P}(B) = \mathbb{P}\left(B|A\right)\mathbb{P}(A)$$

- **Bayes's Theorem** follows:

$$\mathbb{P}\left(A|B\right) = \mathbb{P}\left(B|A\right)\frac{\mathbb{P}(A)}{\mathbb{P}(B)},$$

  which is useful to frame Bayesian inference. If A is a theory (or parameter vector) and B is evidence, our updated (posterior) probability of the theory A depends on the probability of the observed evidence conditional on the theory.

Confusion Matrix

		True Value		
		Positive	Negative	Total
Estimate	Positive	TP	FP	P'=TP+FP
	Negative	FN	TN	N'=FN+TN
	Total	$TP + FN$	$FP + TN$	N

- Sensitivity: $= TP/(TP + FN)$ (good at identifying who has the disease)
- Specificity: $= TN/(TN + FP)$ (good at identifying who doesn't)

Imagine a scientist develops a test for a disease.

- The test has a false positive rate $\mathbb{P}(Test = + \mid Disease = -) = 5\%$.
- The test has a false negative rate $\mathbb{P}(Test = - \mid Disease = +) = 0\%$.

- If 1% of the population has the disease, what is the odds that someone with a positive test has the disease?
- If 0.05% have the disease, what is the odds that someone with a positive test has the disease?

"If you hear hoofbeats, think horses not zebras"

- Conditional expectations are extremely important in this course and econometrics broadly.
- The **conditional expectation** of $g(X)$ given $Y = y$ is

$$\mathbb{E}\left[g(X) \,|Y = y\right] = \int g(x) \frac{f(x, y)}{f_Y(y)} dx$$

- Note that this conditional expectation is a value for a given value y, but **we can also treat conditional expectations as random variables**. In other words, when the conditioning variable Y is a random variable

$$\mathbb{E}\left[g(X) \,|Y\right]$$

is a random variable.

Law of Iterated Expectations

- The **Law of Iterated Expectations** holds that

$$\mathbb{E}\left[\mathbb{E}\left[g\left(X\right)|Y\right]\right] = \mathbb{E}\left[g\left(X\right)\right]$$

- This is the last operator rule, and the one we will use most: "average within groups, then average the group averages."
- Define $\varepsilon = Y - \mathbb{E}\left[Y|X\right]$. What is $\mathbb{E}\left[\varepsilon X\right]$?

Law of Total Variance

- The conditional-variance cousin of iterated expectations:

$$Var\left(Y\right) = \underbrace{\mathbb{E}\left[Var\left(Y|X\right)\right]}_{\text{within-group}} + \underbrace{Var\left(\mathbb{E}\left[Y|X\right]\right)}_{\text{between-group}}$$

- Total variation in wages $=$ variation *within* education groups $+$ variation in average wages *across* education groups.
- This decomposition is R^2, ANOVA, and “how much can X explain?” all at once — we will meet it again in Lecture 2.

Matrix Notation

Matrix notation: the 15-minute version

Full review (with worked examples) is in the self-study linear algebra handout — read it before next class. What you must recognize on sight:

- Data: n observations on k variables is a matrix X ($n \times k$); one observation is a row x_i'; y is $n \times 1$.
- Transpose: $(AB)' = B'A'$. $X'X$ is $k \times k$, square, and symmetric.
- Inverse: $A^{-1}A = I$; $(AB)^{-1} = B^{-1}A^{-1}$. $(X'X)^{-1}$ exists only when no column of X is a linear combination of the others — this fact will come back as multicollinearity in Lecture 2.
- The one formula to start getting comfortable with now:

$$\hat{\beta} = (X'X)^{-1}X'y$$

You will *read* matrix notation constantly this semester; you will almost never manipulate it by hand.

Thanks